versions that predate the stated cutoff. This discrepancy stems from widespread temporal misalignments within large pretraining corpora—for instance, older documents lingering in recent web crawls—and from the incomplete removal of outdated or duplicated content during data processing (Cheng et al., 2024).

This fallacy persists even if we disregard challenges related to the static and historically bound nature of LLM training data—or issues of grounding, embodiment, and subjective experience. As the average human fallacy illustrates, responses from LLMs cannot be assumed *a priori* to represent average responses of the targeted human group. Even when LLMs and humans show alignment, this correlation should not be confused with equivalence in cognitive processes or mechanisms (the alignment as explanation fallacy; **Box 4**). This leads to an epistemic dilemma: Generalizing findings from LLMs to humans requires corroboration with actual human data, undermining the basic premise of the substitution proposition.

Additionally, substitution risks creating closed-loop information systems. When models trained on historical data are used as primary tools for generating new data, they perpetuate a self-referential loop that creates a distorted view of the present by amplifying the past (including its biases, errors, and oversights) rather than reflecting current human thought or behavior. This can lead to misleading statistical inferences reflecting model artifacts rather than genuine psychological phenomena and behavioral patterns.

Even with up-to-date training data, excluding human participants leaves LLMs simulating humans in ways detached from rich, evolving realities. Such detachment can entrench outdated knowledge, weaken the diversity vital for human progress, and create epistemic echo chambers. Thus, LLMs should serve as supplementary rather than primary tools for understanding the human mind.

To mitigate the substitution fallacy, safeguards include using sequential validation with human participants, benchmarking against time-sensitive data, integrating mixed methods, ensuring temporal transparency, and implementing closed-loop detection techniques. Consider several approaches:

1. **Sequential validation**: Implement a sequential research design where LLM explorations are followed by validation with human participants. Use LLMs for hypothesis generation or initial exploration, but validate key findings with relevant human data (Gui & Toubia, 2023).
2. **Benchmarking against time-sensitive data**: Regularly benchmark LLM responses against recent human data to assess temporal drift in model outputs compared to current human attitudes and behaviors. This helps establish the temporal boundaries of generalizability for LLM-based findings (Cheng et al., 2024).
3. **Temporal transparency**: Explicitly document training data cutoff dates and potential temporal limitations in research reports, particularly in rapidly evolving domains (Cheng et al., 2024).

Concluding Remarks

Recent advances in human-level AI are renewing the classic debate on the role of computing artifacts in understanding the human mind and brain (Simon, 1983). This paper critically assesses the emerging proposition of substituting human participants with LLMs in behavioral and social sciences. By exposing six fallacies inherent in this replacement perspective, the analysis underscores that, despite their human-like language production capabilities, current LLMs do not—and as presently conceived, cannot—substitute for the nuanced intricacies of human thought. Unlike the statistical text prediction that drives current LLMs, human intelligence emerges from embodied interaction with the world—grounded in sensory experiences, enriched by multimodal integration, and shaped by subjective consciousness. The predominantly linguistic nature of LLMs further constrains their ability to capture the breadth of human experience, including nonverbal cues, implicit attitudes, and real-world behaviors.

By identifying challenges to research integrity and providing practical guidelines, the analysis supports the simulation perspective: LLMs serve as tools for simulating roles and modeling cognitive processes, complementing but not replacing humans. As outlined in **Box 4**, this perspective helps investigators distinguish between research contexts where output-level simulation suffices (pragmatic applications like rapid prototyping) and those requiring deeper mechanistic evidence (theoretical claims about cognitive processes). In practice, researchers should leverage LLMs primarily for hypothesis generation, theory development, and rapid prototyping—then validate with human participants. This sequential approach capitalizes on model strengths (comprehensive knowledge, efficient simulation) while acknowledging their limitations (lack of grounding, representational biases). Implementing the controls and considerations outlined for each fallacy can substantially improve research quality and interpretability.

As emphasized in **Box 3**, understanding model limitations requires distinguishing between technical and conceptual constraints. While technical limitations may be addressed through engineering advances, conceptual limitations represent fundamental challenges to using LLMs as psychological models. As these technologies evolve, we must continuously re-evaluate their capabilities and limitations, develop appropriate benchmarks, and establish guidelines for responsible integration. This perspective invites us to reconsider the role of AI in behavioral and cognitive science—as a mirror through which we can better understand the similarities and differences between human intelligence and machine intelligence. And the limitations of apparently human-like models in replicating human thought may bring us a deeper appreciation of the complexity and wonder of the mind.

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